《通信原理》第2章

2.7 能量谱密度

概念

- 能量信号$x(t)$的能量谱密度$E_x(f)$是其总能量E_x在频率轴$-\infty < f < \infty$上分布密度。
- 对于任意频率值f_0以及充分小量Δ，$x(t)$的能量落入无穷小区间$[f_0, f_0 + \Delta]$内的数量是$E_x(f_0)\Delta$。
- 总能量是能量密度的积分：

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \mathrm{d}t = \int_{-\infty}^{\infty} |X(f)|^2 \mathrm{d}f$$

- 同理，能量信号$x(t), y(t)$的互能量谱密度$E_{xy}(f)$是其互能量E_{xy}在频率轴$-\infty < f < \infty$上分布密度，其积分是互能量。

$$E_{yx} = \int_{-\infty}^{\infty} y(t)x^*(t)\mathrm{d}t = \int_{-\infty}^{\infty} Y(f)X^*(f)\,\mathrm{d}f$$

- 数学定义：
  - 两个能量信号$x(t), y(t)$的互能量谱密度定义为

$$E_{xy}(f) = X(f)Y^*(f)$$

  - 当$x(t) = y(t)$时称为自能量谱密度（简称能量谱密度）

$$E_x(f) = X(f)X^*(f) = |X(f)|^2$$

- 帕瑟瓦尔定理
  - 能量是能量谱密度的积分

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \mathrm{d}t = \int_{-\infty}^{\infty} |X(f)|^2 \mathrm{d}f$$

  - 互能量是互能量谱密度的积分

$$E_{yx} = \int_{-\infty}^{\infty} y(t)x^*(t)\mathrm{d}t = \int_{-\infty}^{\infty} Y(f)X^*(f)\,\mathrm{d}f$$

- 互相关函数的傅氏变换是互能量谱密度

$$R_{xy}(\tau) \Leftrightarrow X(f)Y^*(f)$$

$$R_{yx}(\tau) \Leftrightarrow Y(f)X^*(f)$$

- 自相关函数的傅氏变换是自能量谱密度

$$R_x(\tau) \Leftrightarrow |X(f)|^2$$

证明： 对$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t+\tau)y^*(t)\,\mathrm{d}t$两边做关于$\tau$的傅氏变换

$$\mathcal{F}\left[R_{xy}(\tau)\right] = \int_{-\infty}^{\infty} X(f)\mathrm{e}^{\mathrm{j}2\pi ft} \cdot y^*(t)\,\mathrm{d}t = X(f)\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{j}2\pi ft} \cdot y^*(t)\,\mathrm{d}t$$

$$= X(f)\left[\int_{-\infty}^{\infty} y(t)\mathrm{e}^{-\mathrm{j}2\pi ft}\,\mathrm{d}t\right]^* = X(f)Y^*(f)$$

证明二：$R_{xy}(\tau)=\int_{-\infty}^{\infty}x(t+\tau)y^{*}(t)\,\mathrm{d}t$是$x(t+\tau)$与$y(t)$的内积。时域内积等于频域内积。$x(t+\tau)$的傅氏变换是$X(f)\mathrm{e}^{\mathrm{j}2\pi f\tau}$。$y(t)$的傅氏变换是$Y(f)$，故

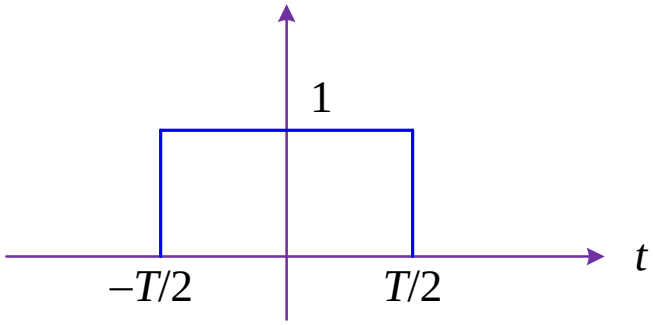
$$
\begin{aligned}
R_{xy}(\tau)&=\int_{-\infty}^{\infty}x(t+\tau)y^{*}(t)\mathrm{d}t\\
&=\int_{-\infty}^{\infty}X(f)\mathrm{e}^{\mathrm{j}2\pi f\tau}\cdot Y^{*}(f)\mathrm{d}f\\
&=\int_{-\infty}^{\infty}X(f)Y^{*}(f)\mathrm{e}^{\mathrm{j}2\pi f\tau}\mathrm{d}f
\end{aligned}
$$

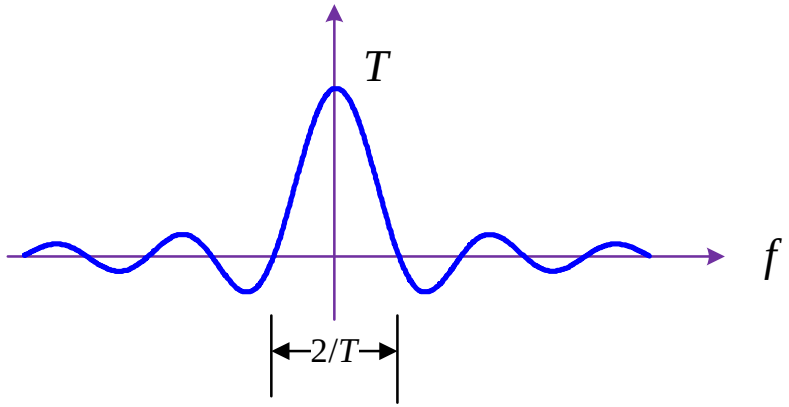
即$R_{xy}(\tau)$是$X(f)Y^{*}(f)$的傅氏反变换

Remark: 注意式中的共轭关系。设有两个信号$x_1(t),x_2(t)$，其傅氏变换分别是$X_1(f),X_2(f)$，则时域内积等于频域内积的关系式为

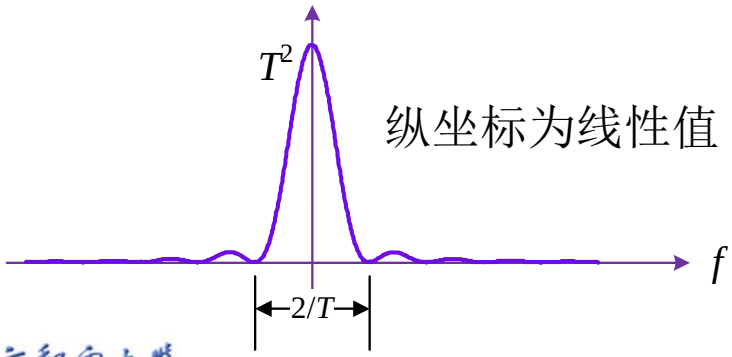
$$\int_{-\infty}^{\infty}x_1(t)x_2^{*}(t)\mathrm{d}t=\int_{-\infty}^{\infty}X_1(f)X_2^{*}(f)\mathrm{d}f$$

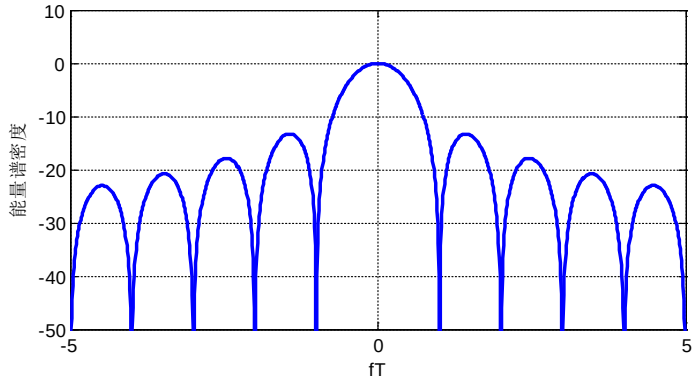
- 例：矩形脉冲$x(t) = \text{rect}\left(\frac{t}{T}\right)$的傅氏变换是$X(f) = \mathcal{F}[x(t)] = T \cdot \text{sinc}\,(fT)$





能量谱密度是$E_x(f) = |X(f)|^2 = T^2 \cdot \text{sinc}^2\,(fT)$



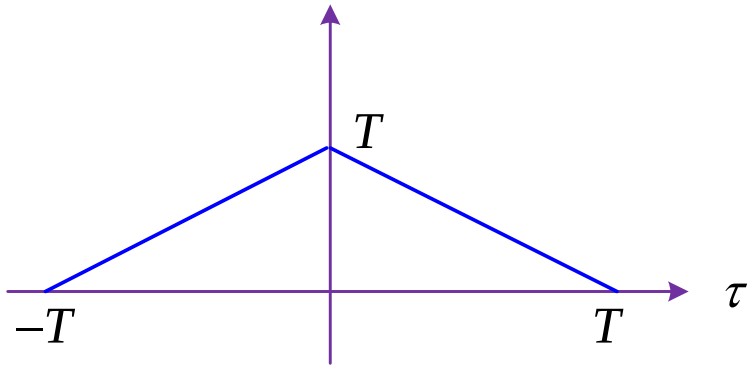


纵坐标为分贝值

北京郵電大學
BEIJING UNIVERSITY OF POSTS AND TELECOMMUNICATIONS

矩形脉冲的自相关函数是

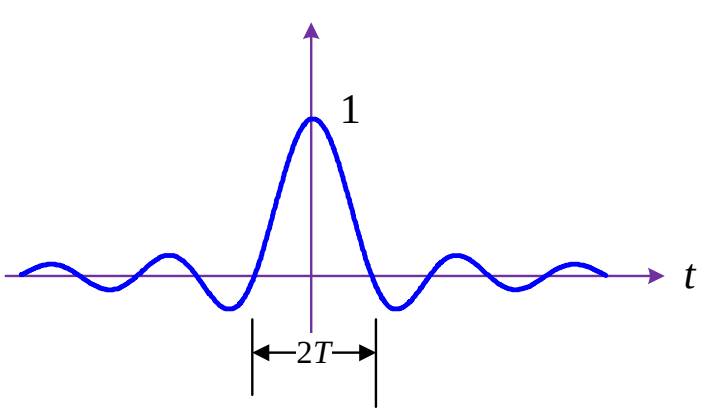
$$R_x(\tau)=\int_{-\infty}^{\infty}x(t)x(t+\tau)\mathrm{d}t=\int_{-T/2}^{T/2}x(t+\tau)\mathrm{d}t=\begin{cases}T+\tau, & -T<\tau<0\\ T-\tau, & 0<\tau<T\\ 0, & |\tau|\geq T\end{cases}$$

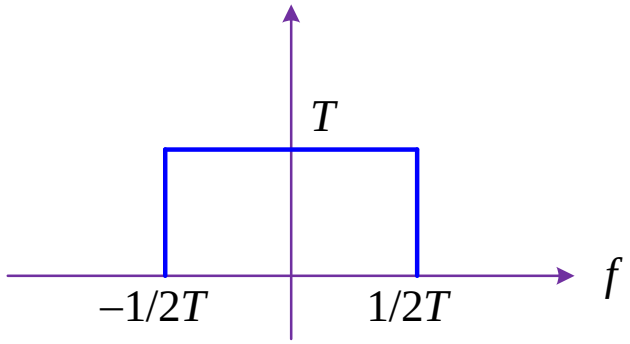


该自相关函数是能量谱密度$|X(f)|^2=T^2\cdot\mathrm{sinc}^2\,(fT)$

的傅氏反变换

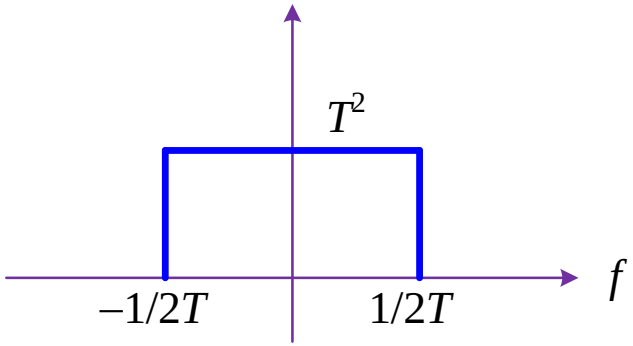
- 例：**sinc脉冲**$x(t)=\mathrm{sinc}\left(\frac{t}{T}\right)$的傅氏变换是$X(f)=T\cdot\mathrm{rect}(fT)$

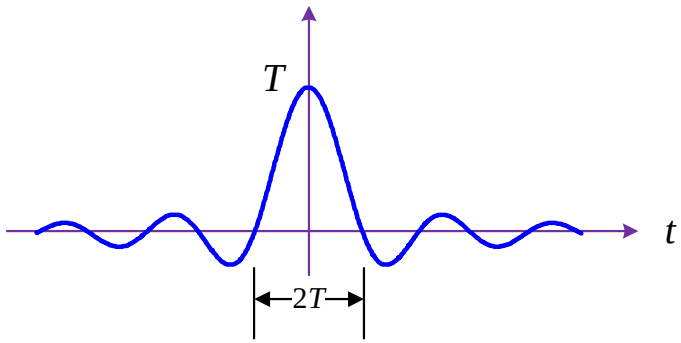




能量谱密度是$E_x(f)=|X(f)|^2=T^2\cdot\mathrm{rect}(fT)$

自相关函数是$R_x(\tau)=\mathcal{F}^{-1}[T^2\cdot\mathrm{rect}(fT)]=T\cdot\mathrm{sinc}\left(\frac{\tau}{T}\right)$





小结

- 互能量谱密度：$E_{xy}(f) = X(f)Y^*(f) \quad E_{yx}(f) = Y(f)X^*(f)$
- 能量谱密度：$E_x(f) = |X(f)|^2$
- 面积是能量：

$$E_{yx} = \int_{-\infty}^{\infty} E_{yx}(f)\,\mathrm{d}f \qquad E_{xy} = \int_{-\infty}^{\infty} E_{xy}(f)\,\mathrm{d}f \qquad E_x = \int_{-\infty}^{\infty} E_x(f)\,\mathrm{d}f$$

- 相关函数与能量谱密度是傅氏变换关系

$$R_{xy}(\tau) \Leftrightarrow X(f)Y^*(f) \qquad R_{yx}(\tau) \Leftrightarrow Y(f)X^*(f)$$